

SL No of QP : 5534
 Unique Paper Code : 2352012401
 Name of the Paper : Sequences and Series of Functions
 Name of the Course : B.Sc. (H) Mathematics
 Semester : IV
 Duration : 3 hours
 Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on the receipt of this question paper.
2. All questions carry equal marks.
3. Attempt any two parts from each question.

1. (a) Define uniform convergence for sequence of functions (f_n) defined on $\mathbb{R}$. Show that a sequence (f_n) of bounded functions on $A \subseteq \mathbb{R}$ converges uniformly on A to f if and only if $\|f_n - f\|_A \rightarrow 0$.

(b) Show that the sequence $f_n(x) = \frac{nx}{1+n^2x^2}$ converges uniformly in the interval $[a, \infty)$ where $a > 0$ but it does not converge uniformly on the interval $[0, \infty)$.

(c) Let $(f_n), (g_n)$ be sequences of bounded functions on A that converge uniformly on A to f, g , respectively. Show that $(f_n g_n)$ converges uniformly on A to fg .

2. (a) Let $f_n(x): [0,1] \rightarrow \mathbb{R}$ be defined for $n \geq 2$ by

$$f_n(x) := \begin{cases} n^2 x & \text{for } 0 \leq x \leq \frac{1}{n} \\ -n^2 \left(x - \frac{2}{n}\right) & \text{for } \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & \text{for } \frac{2}{n} \leq x \leq 1 \end{cases}$$

Evaluate pointwise limit f of f_n and show that (f_n) does not converge uniformly to f on the interval $[0,1]$.

(b) Let (f_n) be a sequence of integrable functions on $[a, b]$ and suppose that (f_n) converges uniformly to f on $[a, b]$. Show that f is integrable on $[a, b]$ and

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

(c) Show that the sequence $f_n(x) = n^2 x^2 e^{-nx}$ converges uniformly on $[a, \infty)$ where $a > 0$ but does not converge uniformly on the interval $[0, \infty)$.

3. (a) State and prove Weierstrass M-test for uniform convergence of series of functions.

Hence, check for uniform convergence of series $\sum_{n=1}^{\infty} \frac{1}{x^2+n^2}$, $x \in \mathbb{R}$.

(b) Discuss the pointwise convergence of the series $\sum_{n=1}^{\infty} \frac{x^n}{x^{n+1}}$, $x \geq 0$. Also, show that the given series is not uniformly convergent on $[0,1)$ but is uniformly convergent on $[0, \frac{1}{2})$.

(c) Prove that if $\sum f_n$ converges uniformly to f on a domain $D \subseteq \mathbb{R}$ and each f_n is continuous on $D \subseteq \mathbb{R}$ to $\mathbb{R}$ then f is continuous on D .

4. (a) Let f_n be a real valued function on $[a, b]$ that has derivative f'_n on $[a, b]$ for each $n \in \mathbb{N}$. Suppose that the series $\sum f_n$ converges for at least one point of $[a, b]$ and the series of derivatives $\sum f'_n$ converges uniformly on $[a, b]$. Show that there exists a real valued function f on $[a, b]$ such that $\sum f_n$ converges uniformly to f on $[a, b]$. Also, that f has a derivative on $[a, b]$ and $f' = \sum f'_n$.

(b) State and prove the Cauchy criterion for the uniform convergence of series of functions $\sum f_n$. Use it to prove the non uniform convergence of $\sum_{n=1}^{\infty} \frac{1}{n^2 x^2}$ for $|x| < 1$.

(c) Prove that the series $\sum_{n=1}^{\infty} \frac{1}{x^{n+1}}$ is convergent for $x > 1$ and divergent for $0 \leq x \leq 1$. Also, show that the series converges uniformly on $[a, \infty)$, $a > 1$.

5. (a) (i) Find the exact interval of convergence for the power series $\sum_{n=1}^{\infty} a_n x^n$ where $a_n = \frac{3^n}{n4^n}$.

(ii) Let $f(x) = |x|$ for $x \in \mathbb{R}$. Is there a power series $\sum_{n=0}^{\infty} a_n x^n$ such that $f(x) = \sum_{n=0}^{\infty} a_n x^n$ for all x ? Justify.

(b) Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $R > 0$. If $0 < R_1 < R$, then show that the power series converges uniformly on $[-R_1, R_1]$ to a continuous function.

(c) Show that $\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$ for $|x| < 1$. Apply Abel's theorem to show that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$.

6. (a) Define logarithm function L on $(0, \infty)$. Show that the function L satisfies the following:

(i) $L'(x) = \frac{1}{x}$ for $x > 0$.

(ii) $L(xy) = L(x) + L(y)$ for $x > 0, y > 0$.

(b) Show that $\sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}$ for $|x| < 1$. Evaluate $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

(c) Show that there does not exist a sequence of polynomials converging uniformly on $\mathbb{R}$ to $f(x) = e^x$.